Hardest Material Tapped (All Aero. or Federal Spec.; Steel or Alloy Number)	Screw Size No.	Thickness of Hardest Material Tapped (Inches)									
		.015 to .018 incl.	.018 to .040 incl.	.041 to .056 incl.	.057 to .071 incl.	.072 to .089 incl.	.091 to .110 incl.	.103 to .115 incl.	.113 to .137 incl.	.132 to .148 incl.	Over .148
		Drill Holes For Screws to Diameters Specified Below (Inches)									
Steel											
Carbon Mild Quarter-Hard (55,000 p.s.i. Min. T. S.) Spec. AN-QQ-S-651	4-40	.086	.086	.086	.093	.093		.096	.096	.096	
	6-32	.104	.104	.104	.113	.113		.113	.120	.120	
	8-32	.128	.128	.128	.140	.140		.140	.144	.144	
	10-24	.149	.149	.149	.154	.154		.154	.161	.161	
	1/4-20	.199	.199	.199	.209	.209		.209	.213	.213	
Chrome-Molybdenum Normalized (1) (90,000 p.s.i. Min. T. S.) Spec. AN-QQ-S-685	4-40	.089	.089	.089	.089	.096	.096		.096		
	6-32	.110	.110	.110	.110	.116	.120		.120		
	8-32	.136	.136	.136	.136	.140	.144		.144		
	10-24	.157	.157	.157	.157	.159	.161		.161		
	1/4-20	.209	.209	.209	.209	.213	.218		.218	.218	
Corrosion Resisting (1) Half Hard (150,000 p.s.i. Min. T. S.)	4-40	.093	.093	.096							
	6-32	.110	.110	.110	.116						
	8-32	.136	.136	.140	.140	.144					
	10-24			.159	.161	.161	.166				
	1/4-20			.205	.205	.209	.213	.218	.221		
Aluminum Alloy											
Alum.-Magnesium Chrom. Half-Hard (34,000 p.s.i. Min. T. S.) Spec. QQ-A-318	4-40	.086	.086	.086	.086	.089			.093		
	6-32	.104	.104	.104	.104	.106			.106		
	8-32	.128	.128	.128	.128	.136			.140		
	10-24	.149	.149	.149	.149	.152			.154		
	1/4-20	.193	.193	.199	.199	.204			.204		
Alum.-Copper-Magn.-Manganese Heat Treated (52,000 p.s.i. Min. T. S.) Spec. QQ-A-361	4-40	.086	.086	.086	.086	.089	.089		.093		
	6-32	.104	.104	.104	.104	.106	.106		.106		
	8-32	.128	.128	.128	.128	.136	.140		.140		
	10-24	.149	.149	.149	.149	.152	.152		.154		
	1/4-20	.199	.199	.199	.199	.204	.204		.204		
Alum.-Copper-Magn. (1.5%) Manganese Heat Treated (62,000 p.s.i. Min. T. S.) Spec. QQ-A-355	4-40	.086	.086	.086	.089	.089					
	6-32	.104	.104	.104	.109	.109					
	8-32	.129	.129	.129	.136	.136					
	10-24	.149	.149	.149	.152	.152					
	1/4-20	.199	.199	.199	.203	.203					
Castings (Aluminum Alloy or Aluminum)	4-40				.096	.096	.096	.096	.098	.098	.098
	6-32				.116	.116	.116	.116	.120	.120	.120
	8-32				.144	.144	.144	.144	.147	.147	.147
	10-24				.161	.161	.161	.161	.166	.166	.166
	1/4-20				.221	.221	.221	.221	.228	.228	.228

(1) LUBRICATE WITH HEAVY CUTTING OIL.

② CANCELED AFTER USE MS

① ABOVE HOLE SIZES SHALL APPLY ONLY TO SELF TAPPING SCREWS CONFORMING TO DRAWINGS AN504 AND AN506.

DRILL EACH PLY OR PLATE OF THE ASSEMBLY TO THE SIZE STATED FOR ITS HARDNESS AND THICKNESS. FOR ASSEMBLIES ALREADY RIVETED TOGETHER, THE SIZE OF THE HOLE SHALL DEPEND ON THE HARDEST PLY OR PLATE EMPLOYED AS A NUT PLATE OR A TAPPED INTERMEDIATE PLATE.

WHENEVER PRACTICABLE THE DIAMETER OF THE SCREW SHALL BE SUCH THAT THE DRILL HOLE SIZE WILL FALL BETWEEN THE VERTICAL HEAVY LINES IN THE TABLE.

APPROVED 10-3-42 REVISED ① 4-12-43 ②

ARMY-NAVY AERONAUTICAL DESIGN STANDARD

SELF TAPPING SCREWS - MACHINE SCREW THREAD (HOLE SIZES FOR)

AND10326

NOT A PART NUMBER

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